

International Marketing Analytics

Forecasting and time series

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

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Today

Concepts | By hand | How good? | Real data | Python | In class

Theory, Parts 1 to 3

- Trend, season, cycle; stationary or not?
- Simple forecasts and smoothing by hand
- How good is a forecast?

Practice, Parts 4 to 6

- Real data: clothing sales in Australia
- Python: polars, plotnine, statsmodels
- Online quiz and its answers
- Coding exercise and Posit Assistant

Question of the day: how much will we sell next month, and how sure can we be?

Our example: planning demand in two shops

The problem: a marketing manager must plan before demand arrives.

The question: how much will we sell next week, next quarter, next Christmas?

Stock

order before the peak

Staff

shifts for the busy weeks

Media budget

spend when buyers are there

Two tiny shops first: a phone shop (contracts per week) and a chocolate web shop (sales per quarter); real data in Part 4.

Invented round numbers (hand_examples.csv), built for this course

The data: one row per period

Phone shop: new contracts per week

Week	1	2	3	4	5	6	7	8
Contracts	40	44	42	46	50	49	53	54

Chocolate shop: sales per quarter, k EUR

	Q1	Q2	Q3	Q4
Year 1	20	16	20	30
Year 2	24	20	23	35
Year 3	28	24	26	40

period: the time step: week 1 to 8, quarter 1 to 12

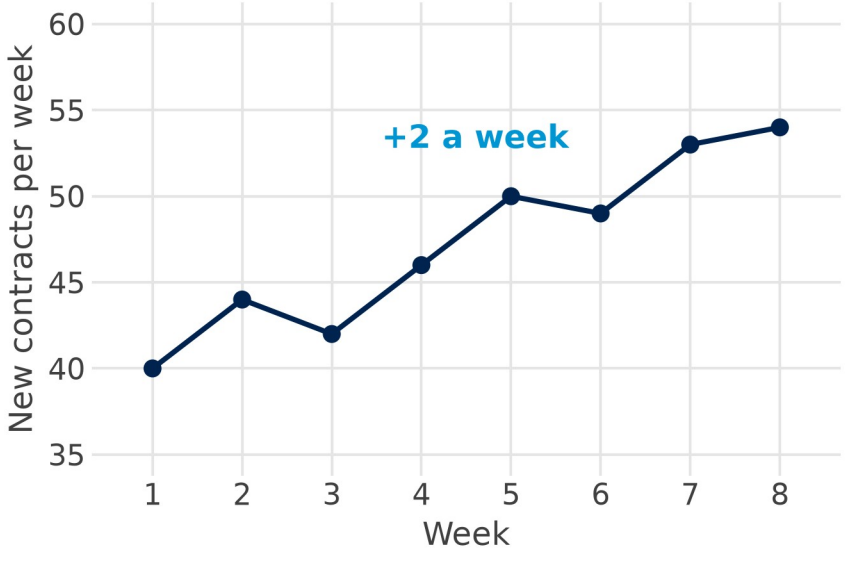
value: what we forecast

rules: time order, equal steps, no gaps

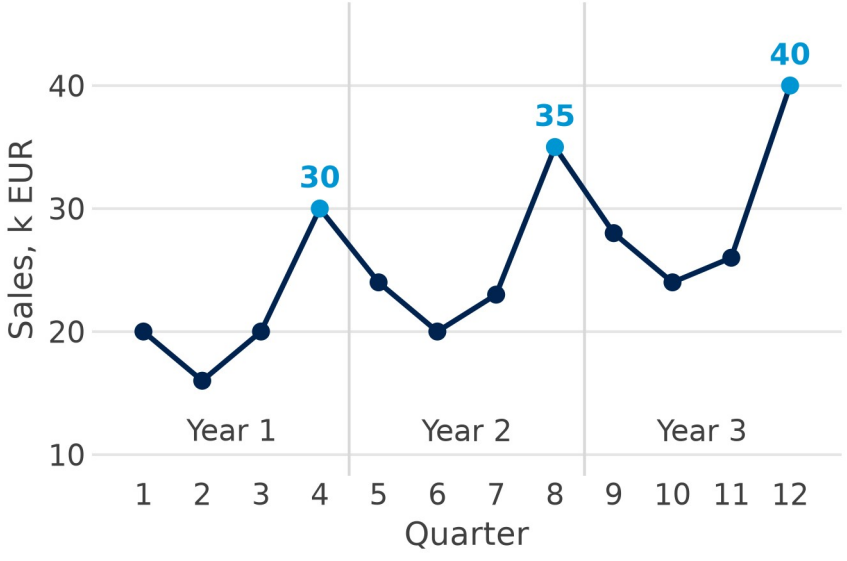
File: `data/forecasting/hand_examples.csv` invented, 20 rows: example, period, value

Invented round numbers (hand_examples.csv)

First look: plot the series



Trend: up about 2 a week



Trend and season: every Q4 peaks

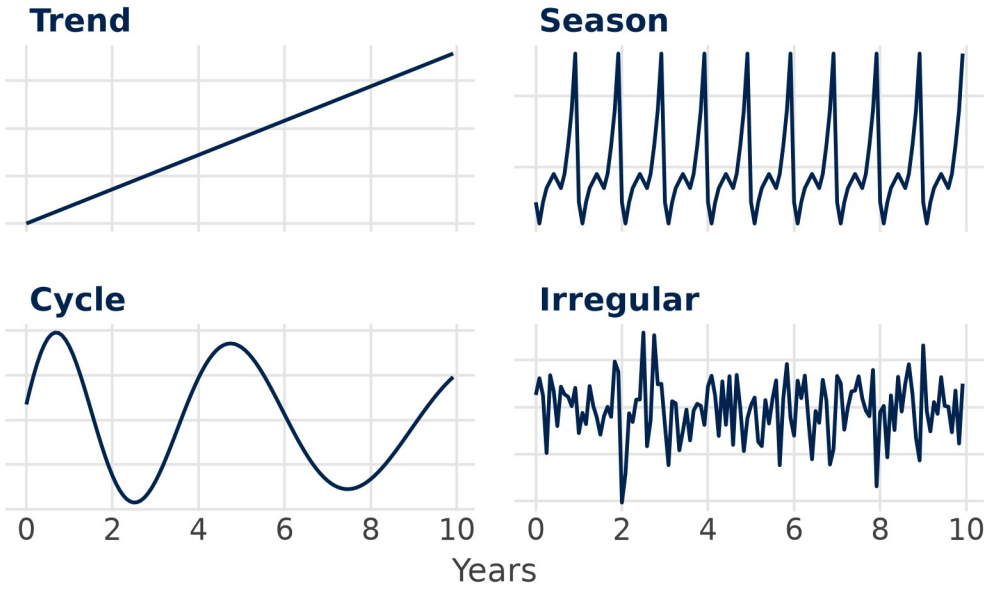
Plot first: the eye finds trend and season before any model.

Invented round numbers (hand_examples.csv)

Time series concepts

Part 1 of 6 · trend, season, cycle, stationarity

Trend, season, cycle and irregular



Trend: the long-run direction

Season: repeats every year, fixed length (Christmas)

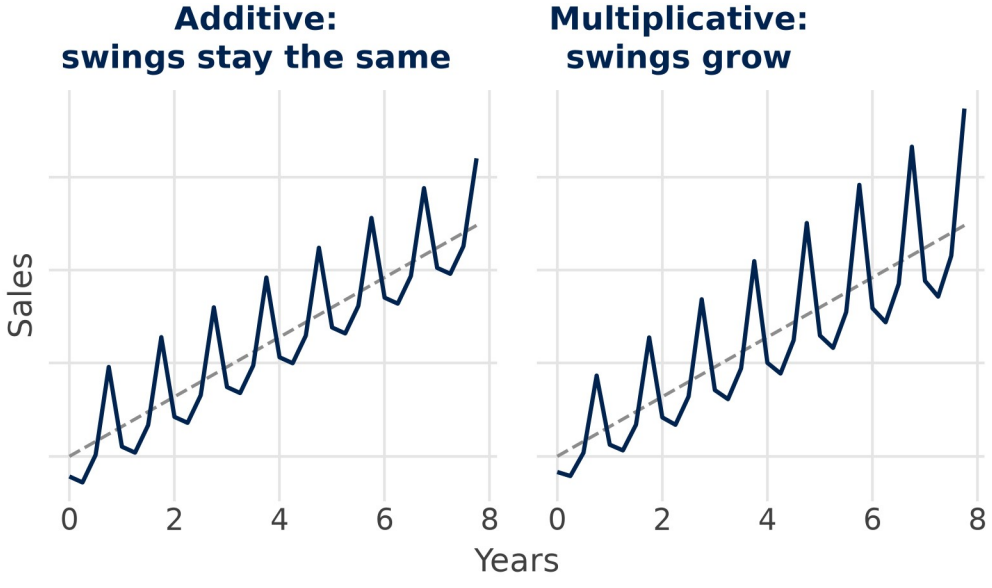
Cycle: ups and downs over years, no fixed length (the economy)

Irregular: what is left: noise, one-off events

Series = trend + season + cycle + irregular (or ×, slide 8): a forecast extends the first three.

Simulated series for illustration

Additive or multiplicative season?



+9.5

chocolate shop: Q4 about 9.5 k EUR above the year mean on average (8.5 to 10.5): additive

× 1.59

clothing (Part 4): December 59 % above an average month, and the swing grows: multiplicative

Multiplicative season: use a multiplicative model or take logs, which makes it additive.

Simulated series for illustration; chocolate shop: invented; clothing: ABS Retail Trade (8501.0), CC BY 4.0

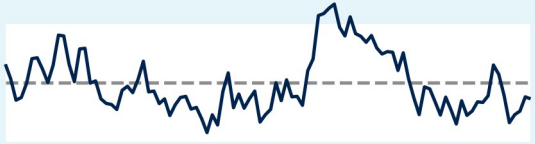
What does stationary mean?

Stationary: the level and the ups and downs stay the same over time; any stretch of the series looks like any other stretch.

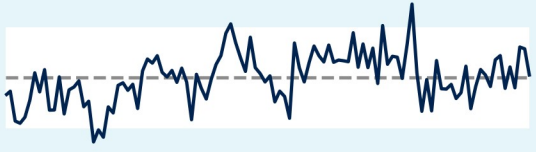
Stationary
noise around a fixed mean



Stationary
wobbles, returns to its mean



Stationary
clothing: change on last year



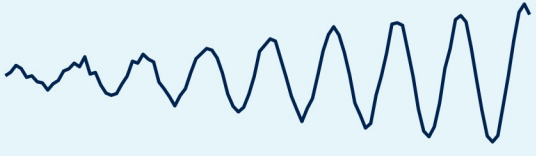
Not stationary
a trend: the level rises



Not stationary
a random walk wanders off



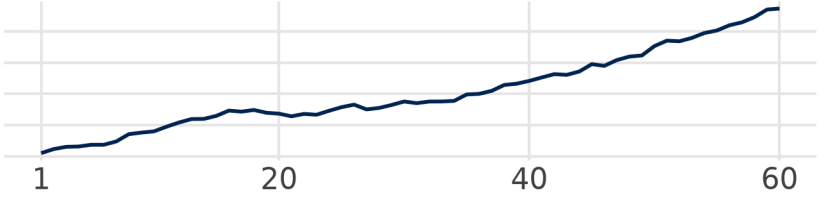
Not stationary
a season whose swings grow



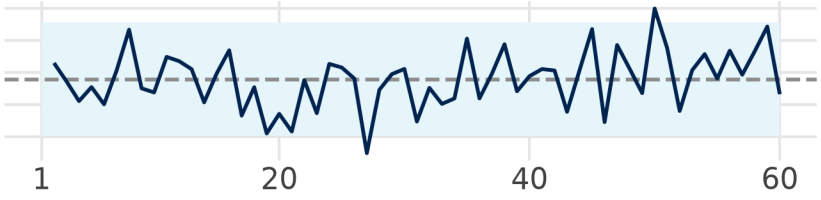
Simulated series for illustration; clothing: ABS Retail Trade (8501.0), CC BY 4.0

Not stationary? Differencing is the fix

A random walk with drift: it rises and wanders



Its changes (this period − the last): stationary



Period

Week	1	2	3	4	5	6	7	8	Average
Contracts	40	44	42	46	50	49	53	54	
Change	–	+4	−2	+4	+4	−1	+4	+1	+2

Differencing: this period − the last; for a season: minus the same period last year. ARIMA-type models need it.

Simulated series for illustration; phone shop: invented round numbers

Simple forecasts by hand

Part 2 of 6 · five methods and smoothing

No change and same change: naive and drift

Week	Contracts	Naive	Error = change
1	40	–	–
2	44	40	+4
3	42	44	−2
4	46	42	+4
5	50	46	+4
6	49	50	−1
7	53	49	+4
8	54	53	+1

54

naive for week 9:
= week 8

56

drift for week 9:
54 + the mean error
(54 − 40) / 7 = +2

Naive's errors are the weekly changes; drift adds their average, +2.

Invented round numbers (hand_examples.csv)

Averages: the mean and a moving average

Week	Contracts	Calculation	Forecast	Error
1	40	–	–	–
2	44	–	–	–
3	42	–	–	–
4	46	(40 + 44 + 42) / 3	42	+4
5	50	(44 + 42 + 46) / 3	44	+6
6	49	(42 + 46 + 50) / 3	46	+3
7	53	(46 + 50 + 49) / 3	48.33	+4.67
8	54	(50 + 49 + 53) / 3	50.67	+3.33

47.25 **mean:** 378 / 8

52 **week 9:** moving average of weeks 6 to 8

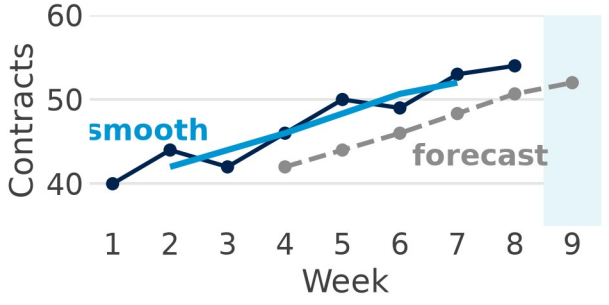
Sum: 40 + 44 + 42 + 46 + 50 + 49 + 53 + 54 = 378

Averages lag behind a trend: every error of the moving average is positive, the forecast is too low.

Invented round numbers (hand_examples.csv)

Rolling averages: smooth or forecast?

Week	Contracts	Calculation	Smooth	Forecast
1	40	–	–	–
2	44	(40 + 44 + 42) / 3	**42**	–
3	42	(44 + 42 + 46) / 3	44	–
4	46	(42 + 46 + 50) / 3	46	**42**
5	50	(46 + 50 + 49) / 3	48.33	44
6	49	(50 + 49 + 53) / 3	50.67	46
7	53	(49 + 53 + 54) / 3	52	48.33
8	54	–	–	50.67



Smooth: the week before, the week and the week after

Forecast: only the 3 weeks before (slide 13)

Same averages, different place: centre the window to see the trend; look only back to forecast.

Invented round numbers (hand_examples.csv)

Exponential smoothing, step by step

Week	Contracts	Level	Calculation (α 0.5)
1	40	40	start: the first value
2	44	42	0.5 × 44 + 0.5 × 40
3	42	42	0.5 × 42 + 0.5 × 42
4	46	44	0.5 × 46 + 0.5 × 42
5	50	47	0.5 × 50 + 0.5 × 44
6	49	48	0.5 × 49 + 0.5 × 47
7	53	50.5	0.5 × 53 + 0.5 × 48
8	**54**	**52.25**	**0.5 × 54 + 0.5 × 50.5**

Rule

```
new level = α × value
          + (1 − α) × old level
forecast = last level
```

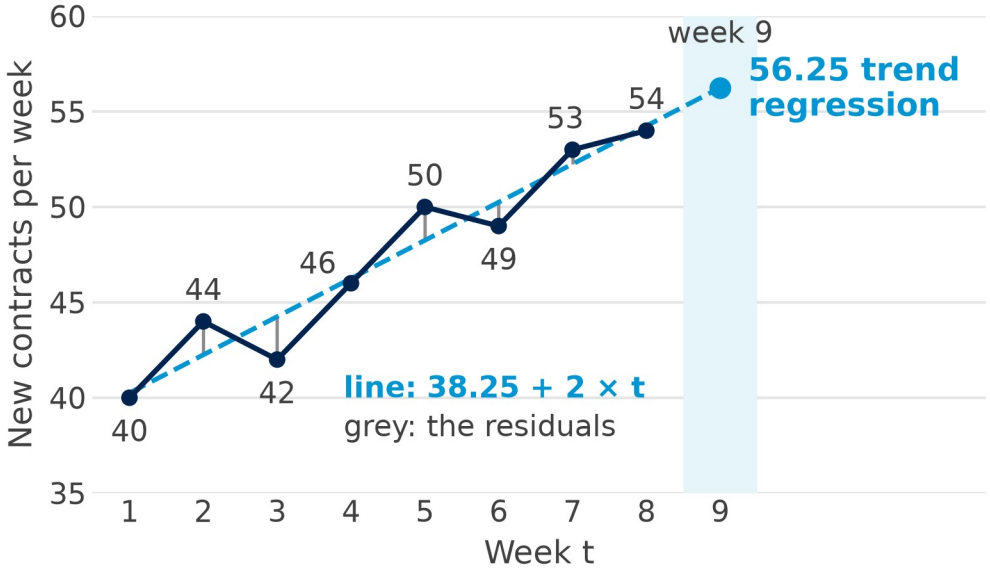
52.25 **forecast** for week 9 (α 0.5)

47.89 **α 0.2:** slower
α 1 = naive (54)

Recent weeks count most: α near 1 follows the data fast, α near 0 averages.

Invented round numbers (hand_examples.csv)

Regression on time: the trend line



By hand

$\bar{t} = 4.5 \quad \bar{y} = 47.25$

$b1 = \Sigma(t - \bar{t})(y - \bar{y}) \,/\, \Sigma(t - \bar{t})^2$

$= 84 \,/\, 42 = 2$

$b0 = \bar{y} - b1 \times \bar{t}$

$= 47.25 - 2 \times 4.5 = 38.25$

week 9: $38.25 + 2 \times 9 = 56.25$

0.926

R^2 = 1 − 13.5 / 181.5
the line fits well

The trend line uses every week: it assumes the trend goes on – and that the errors are independent (slide 32).

Invented round numbers (hand_examples.csv)

The trend line by hand, week by week

Week t	Contracts y	$t - \bar{t}$	$y - \bar{y}$	$(t - \bar{t})(y - \bar{y})$	$(t - \bar{t})^2$	Fitted	Residual
1	40	−3.5	−7.25	+25.375	12.25	40.25	−0.25
2	44	−2.5	−3.25	+8.125	6.25	42.25	+1.75
3	42	−1.5	−5.25	+7.875	2.25	44.25	−2.25
4	46	−0.5	−1.25	+0.625	0.25	46.25	−0.25
5	50	+0.5	+2.75	+1.375	0.25	48.25	+1.75
6	49	+1.5	+1.75	+2.625	2.25	50.25	−1.25
7	53	+2.5	+5.75	+14.375	6.25	52.25	+0.75
8	54	+3.5	+6.75	+23.625	12.25	54.25	−0.25
Sum	**378**	**0**	**0**	**84**	**42**	**378**	**0**

Means: $\bar{t} = 4.5$, $\bar{y} = 47.25$. Fitted = 38.25 + 2 × t; residual = y − fitted.

b1 = 84 / 42 = 2, b0 = 47.25 − 2 × 4.5 = 38.25: two sums give the trend line.

Invented round numbers (hand_examples.csv)

Season by hand: the chocolate shop

Sales, k EUR	Q1	Q2	Q3	Q4	Year mean
Year 1	20	16	20	**30**	21.5
Year 2	24	20	23	**35**	25.5
Year 3	28	24	26	**40**	29.5
Season: average gap	**−1.5**	**−5.5**	**−2.5**	**+9.5**	**sum 0**

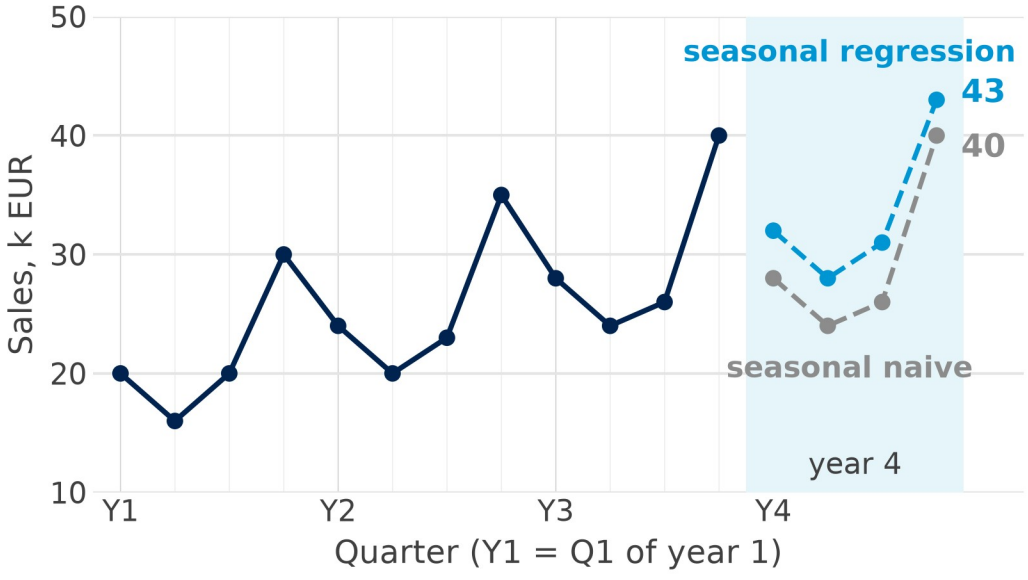
Q4: gaps 30 − 21.5 = +8.5, then +9.5, +10.5
average (8.5 + 9.5 + 10.5) / 3 = +9.5

Trend: year means 21.5 → 25.5 → 29.5
+4 a year, about +1 a quarter (slide 19)

Season = the average gap to the year mean: Q4 sells 9.5 k EUR more than an average quarter.

Invented round numbers (hand_examples.csv)

Two seasonal forecasts for year 4



40 **seasonal naive:**
Q4 of year 4
= Q4 of year 3

43 **seasonal regression:**
$19 + 1 \times 16 + 8$
the trend at t = 16 plus the Q4 effect

Regression, R^2 0.992:
$19 + 1 \times t - 5\ Q2 - 3\ Q3 + 8\ Q4$
Q2, Q3, Q4 = 1 in that quarter, else 0
(Q1 = base)

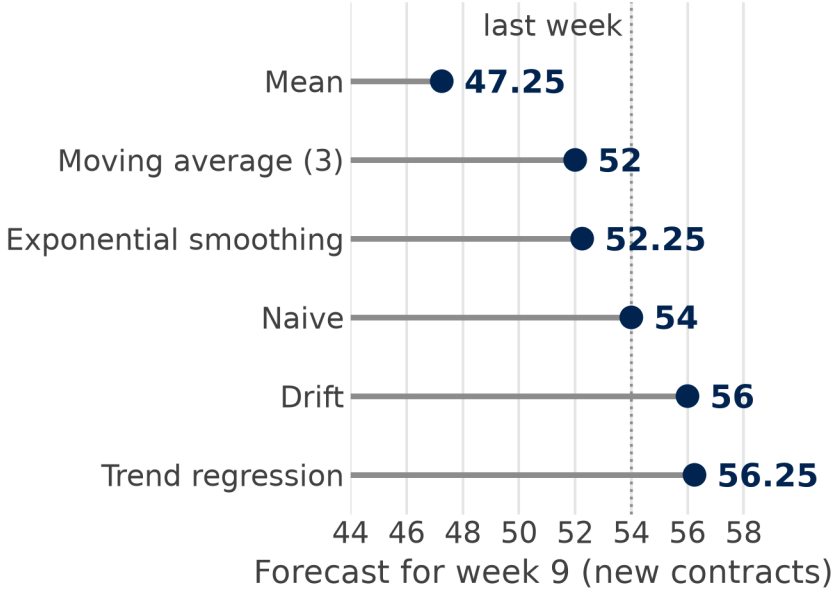
Seasonal naive repeats last year; the regression adds the trend of +1 a quarter.

Invented round numbers (hand_examples.csv)

How good is a forecast?

Part 3 of 6 · hold out, measure, compare

Six forecasts for week 9: which to trust?



1	2	3	4	5	6	7	8
40	44	42	46	50	49	53	54

training period **holdout: weeks 6–8**

Step 1: fit each method on weeks 1–5

Step 2: forecast weeks 6–8, compare with 49, 53, 54

Never at random: test weeks come last.

Six methods, six answers, 9 contracts apart: test each on weeks it has not seen.

Invented round numbers (hand_examples.csv)

Errors and three measures, by hand

Week	Actual	Naive	Error	|Error|	Error²	|Error| / actual
Week 6	49	50	−1	1	1	2.0 %
Week 7	53	50	+3	3	9	5.7 %
Week 8	54	50	+4	4	16	7.4 %
Mean				**2.67**	**8.67**	**5.0 %**

Naive from week 5: 50 for weeks 6, 7 and 8

2.67 **MAE:** mean of |error|

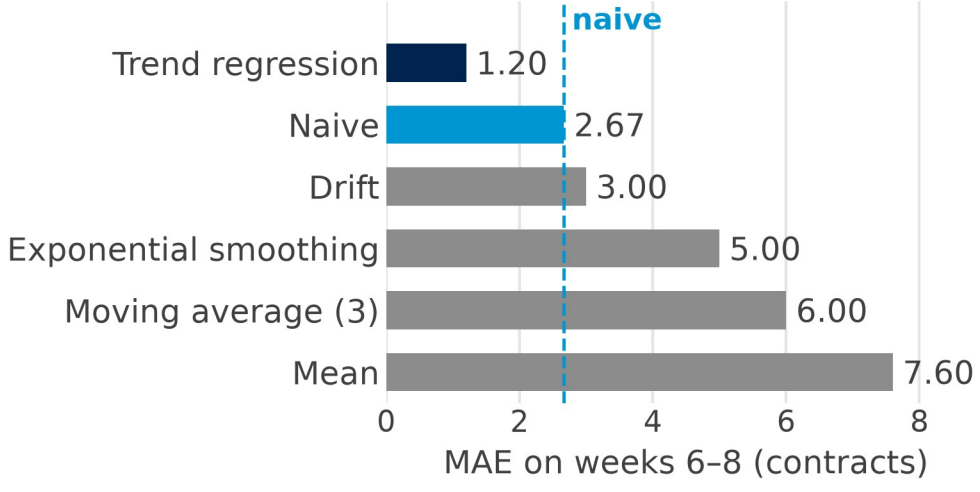
2.94 **RMSE:** √8.67, big errors count more

5.0 % **MAPE:** mean of |error| / actual

Error = actual − forecast: a plus means the forecast was too low.

Invented round numbers (hand_examples.csv)

The ranking: beat the simplest benchmark



1.20 **trend regression:** the only method that beats naive (2.67)

4.00 **chocolate shop, year 3 (MAE, k EUR):** seasonal naive 4.00 vs naive 8.00, so seasonal data need a seasonal benchmark

Forecasts for weeks 6–8, fitted on weeks 1–5: trend regression 37.8 + 2.2 × week = 51, 53.2, 55.4; naive 50; drift 52.5, 55, 57.5; exponential smoothing 47; moving average (3) 46; mean 44.4. RMSE and MAPE give the same order.

Simplest benchmark first: naive, or seasonal naive for seasonal data; keep what beats it.

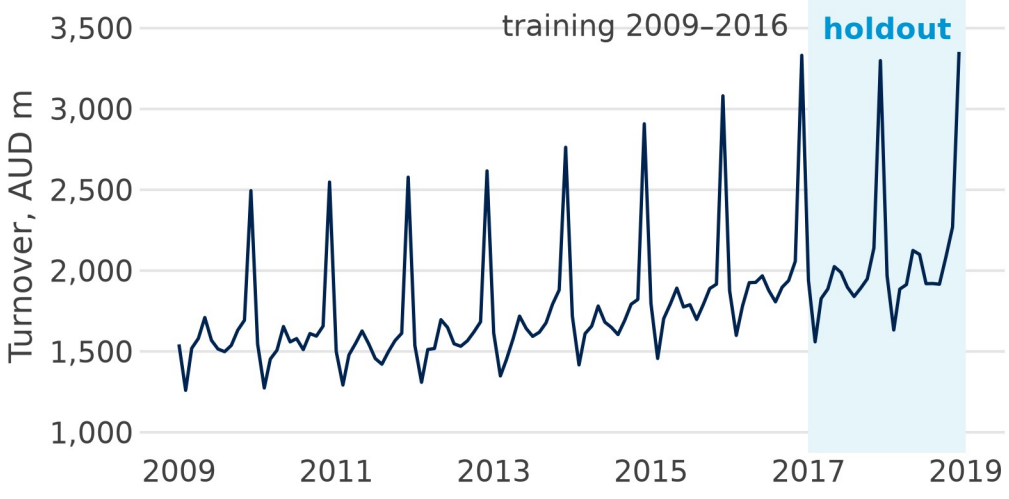
Invented round numbers (hand_examples.csv)

Real data: clothing sales in Australia

Part 4 of 6 · trend, season, holdout, 2019

The data: monthly clothing sales, 2009–2018

The task: a fashion retailer plans stock for 2019 from the market's monthly sales.



Source: Australian Bureau of Statistics (ABS), Retail Trade, open licence (CC BY 4.0)

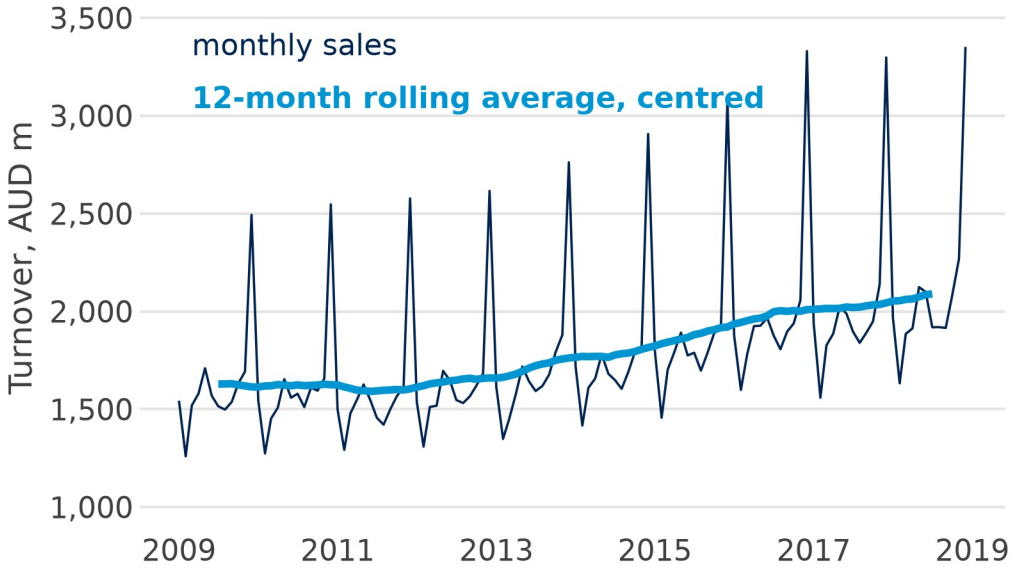
Unit: turnover, AUD million a month, not seasonally adjusted

Period: Jan 2009 to Dec 2018, 120 months: 96 to fit, 24 holdout

File: `data/forecasting/aus_clothing_retail.csv` one row per month: month, value

Source: ABS, Retail Trade, Australia (8501.0), Table 11, clothing, footwear and personal accessories; CC BY 4.0

A 12-month rolling average shows the trend



1,629 — **Jul 2009:** the average month of 2009 (19,546 / 12)

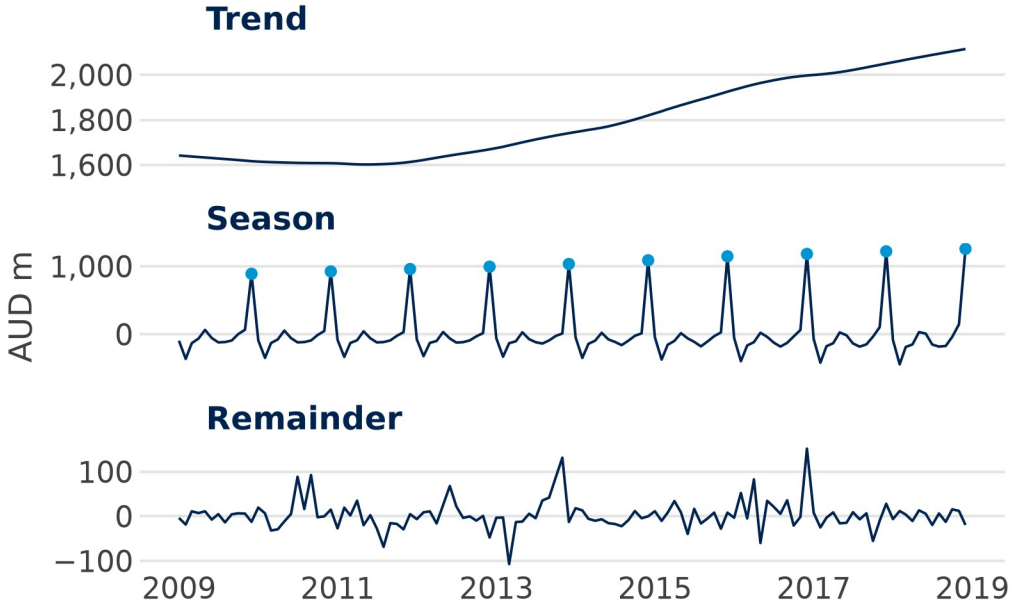
2,090 — **Jul 2018:** the average month of 2018 (25,077 / 12): +28 %

1,591 — **lowest, Jun 2011:** flat to 2012, then growth: the cycle

Twelve months around each month cancel the season: STL (next slide) is a smarter version.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Trend, season and cycle in the real series



× 1.59 — **December** vs an average month; February × 0.80

+28 % — **2009 to 2018;** flat to 2012 (a cycle), then about +4 % a year

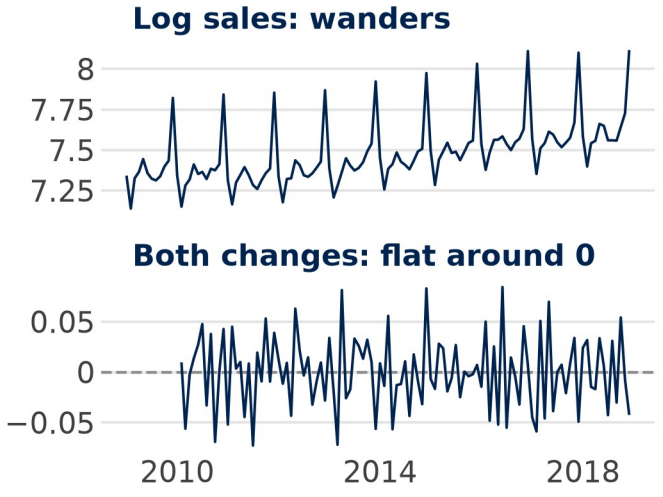
1,257 — **December swing** in 2018, AUD m (2009: 890): it grows with the level

Strong trend and season: a good method must model both.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Is it stationary? The ADF test

Series	ADF p	Stationary?
Sales (levels)	0.992	no
Log sales	0.987	no
Change on last month	0.011	yes (season stays)
Change on last year	0.003	yes
Both changes, log sales	**< 0.001**	**yes**



ADF in plain words: does it wander without a fixed mean? Small p (below 0.05): no, stationary.

Difference until it is stationary: SARIMA does it for us (next slide).

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Two models that learn trend and season

Holt-Winters / ETS

- Smooths level, trend and season (multiplicative)
- Each has its own weight α, β, γ (slide 15)
- ETS: the same idea as a statistical model, so it also gives bands

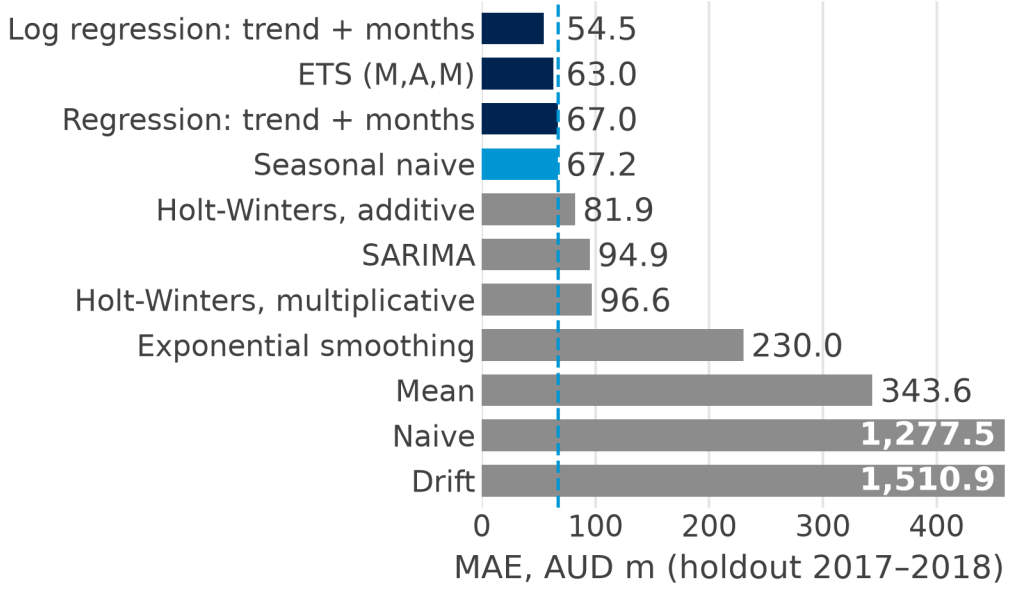
SARIMA

- Differences remove trend and season
- Then it models the memory: how much of last month's and last year's surprise carries on
- Ours: SARIMA$(0,1,1)(0,1,1)_{12}$ on log sales

log sales → change on last month → change on last year → memory of surprises

Both learn trend and season from the data; the holdout decides which forecasts better.

Holdout 2017–2018: beat the seasonal naive



3 of the other 14 methods beat the **seasonal naive** (67.2)

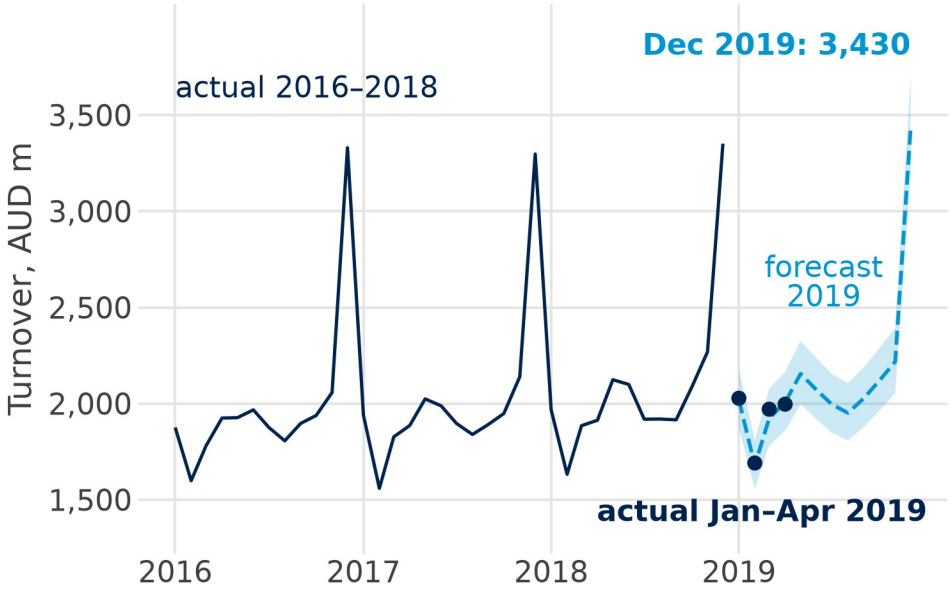
54.5 **best:** log regression, trend + months (MAPE 2.5 %)

3,331 **December trap:** naive repeats Dec 2016 for 24 months (MAE 1,277.5)

On seasonal data the seasonal naive is the benchmark: only three methods beat it.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0; holdout 2017–2018

The 2019 forecast with a 95 % band



3,430 — **December 2019;** 95 % band 3,178 to 3,702: should hold the actual in 95 of 100 cases

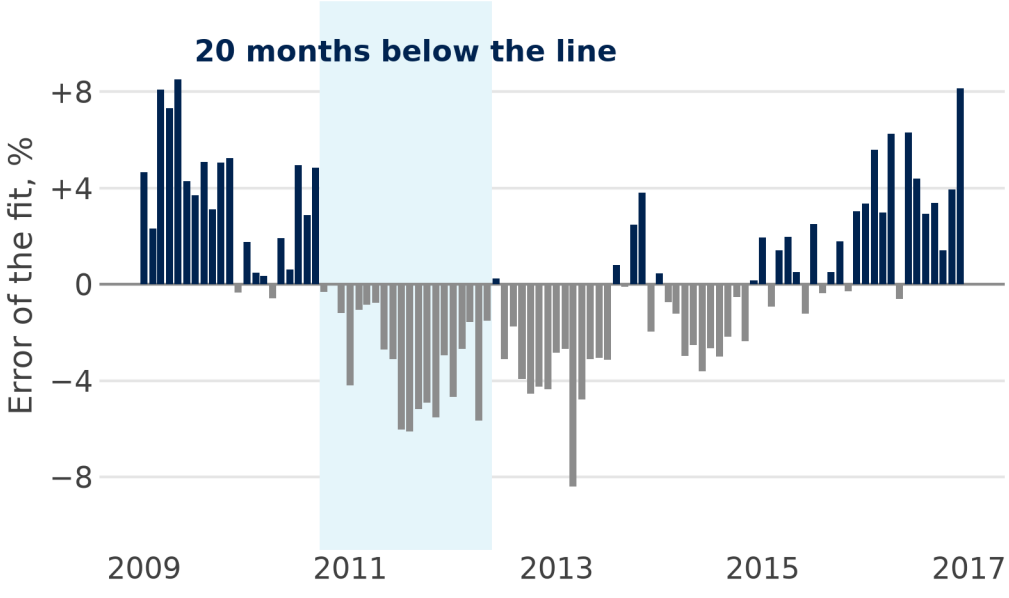
1.0 % — **error Jan–Apr 2019** (MAPE); all 4 inside the band

25,595 — **forecast for 2019** (2018: 25,077)

Refit the winner on all 120 months, then report the forecast with its band.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0

Autocorrelation: errors that stick together



0.72 **autocorrelation:** this month's error is like last month's (0 = none)

0.49 **Durbin–Watson:** 2 = no autocorrelation

524 **December band width,** AUD m (regression); SARIMA's widens from 209 in Jan to 589

Errors that stick together: the regression band does not widen with the horizon; for later months use the SARIMA band.

Source: ABS Retail Trade (8501.0), clothing; CC BY 4.0; training months 2009–2016

Forecasting in Python



Part 5 of 6 · polars, plotnine, statsmodels

The packages and functions

Task	Package	Function
Load and filter	polars	`pl.read_csv(try_parse_dates=True); filter()`
Plot the series	plotnine	`ggplot(); geom_line()`
Smooth	polars	`rolling_mean(window_size=12, center=True)`
Decompose	statsmodels	`tsa.STL(y, period=12, robust=True).fit()`
Test stationarity	statsmodels	`tsa.adfuller(y)`
Holt-Winters	statsmodels	`tsa.ExponentialSmoothing().fit()`
SARIMA	statsmodels	`tsa.SARIMAX().fit()`
Trend + month regression	statsmodels	`smf.ols("np.log(value) ~ t + C(m)")`
Accuracy and tables	polars, great_tables	`select(pl.all()); mean(); GT()`

Step 1: load, split and plot

Python: polars, plotnine

```
import polars as pl
from plotnine import aes, geom_line, ggplot

df = (pl.read_csv("../../data/forecasting/"
                  "aus_clothing_retail.csv",
                  try_parse_dates=True)
      .filter(pl.col("month").dt.year()
              .is_between(2009, 2018)))
y = df["value"].to_numpy()
train, test = y[:-24], y[-24:]   # 2017-18

(ggplot(df, aes("month", "value"))
 + geom_line())
```

Part	Months	Period
All	120	2009–2018
Train	96	2009–2016
Test	24	2017–2018

The last 24 months are the holdout

Training: 96 months, 2009-01 to 2016-12
Holdout: 24 months, 2017-01 to 2018-12

Look first: trend, a December peak every year, swings that grow.

Step 2: Holt-Winters and SARIMA

Python: statsmodels

```
import numpy as np
import statsmodels.tsa.api as tsa

hw = tsa.ExponentialSmoothing(
    train, trend="add", seasonal="mul",
    seasonal_periods=12,
    initialization_method="estimated").fit()
sar = tsa.SARIMAX(
    np.log(train), order=(0, 1, 1),
    seasonal_order=(0, 1, 1, 12),
).fit(disp=0)

fc = {"snaive": np.resize(train[-12:], 24),
      "hw": hw.forecast(24),
      "sarima": np.exp(sar.forecast(24))}
```

Model	Term	Value
Holt-Winters	α level	0.156
	β trend	0.090
	γ season	0.000
SARIMA	θ month	−0.525
	Θ year	−0.713

Weights and terms fitted on 2009–2016

Forecast 24 months: SARIMA works on log sales; np.exp() brings it back.

Step 3: regression and accuracy

Python: statsmodels, polars

```
import statsmodels.formula.api as smf

d = df.with_columns(
    t=pl.int_range(1, pl.len() + 1),
    m=pl.col("month").dt.month()).to_pandas()
reg = smf.ols("np.log(value) ~ t + C(m)",
              data=d[:-24]).fit()
fc["reg_log"] = np.exp(reg.predict(d[-24:]))

f = pl.DataFrame(fc)  # a column per method
a = f.select((pl.all() - test).abs())
rmse = a.select((pl.all() ** 2).mean().sqrt())
mape = a.select(pl.all() / test).mean() * 100
acc = pl.concat([a.mean(), rmse, mape])
acc.select(pl.all().round(1))  # MAE RMSE MAPE
```

Method	MAE	RMSE	MAPE
Log regression	54.5	69.3	2.5 %
Seasonal naive	67.2	88.2	3.3 %
SARIMA	94.9	107.7	4.7 %
Holt-Winters	96.6	110.7	4.7 %

Holdout 2017–2018, AUD m

Same ranking as slide 30: the log regression wins; the qmd adds ETS, the bands and 2019.

In-class exercise

Part 6 of 6 · quiz, exercise, Posit Assistant

Online quiz: forecasting in five questions

1 Sales: 50, 54, 58. The drift (same change) forecast for the next period is …
A 58 B 62 C 54

2 Exponential smoothing with α 0.5: level 40, new value 50. The new level is …
A 45 B 50 C 42

3 Holdout errors: −2, +2, +4. The MAE is …
A 1.33 B 4.00 C 2.67

4 Monthly sales with a December peak. The benchmark to beat is …
A naive (last month) B seasonal naive (same month last year) C the mean

5 ADF test on sales levels: p 0.99. The series is …
A stationary B white noise C not stationary: difference it

3 minutes alone, then compare with your neighbour.

Answers: in class

Your turn: forecast café sales in Python

Pairs: 35 minutes	Plenary: 5 minutes	Posit Assistant: 10 minutes

1. **By hand:** café bookings 20 … 31: five forecasts for week 7, MAE on weeks 5–6 (Steps 4, 5)
2. **Look:** plot 2009–2018; trend, season, ADF (Steps 1, 6)
3. **Test:** five methods on the 2017–2018 holdout: MAE, RMSE, MAPE (Steps 2, 3)
4. **Forecast:** 2019 with the winner and SARIMA, 95 % bands; check Jan–Apr (Steps 8, 9)

Deliverable: one slide per pair: the 2019 forecast and two sentences.

File: `sessions/01-foundations/forecasting_exercise.qmd` (reuse: forecasting.qmd)

Then: the same tasks with Posit Assistant

How: code first, then open Posit Assistant, type @forecasting_exercise.qmd, one task per prompt

Prompt for task 1

```
@forecasting_exercise.qmd Task 1. Café
bookings 20, 24, 22, 26, 27, 31: naive,
drift, mean, moving average (3) and SES
(α 0.5) for week 7, step by step.
```

Prompt for task 2

```
Task 2. Plot aus_cafes_retail.csv for
2009-2018 with plotnine. Describe trend
and season; run the ADF test on levels
and on the first difference.
```

Prompt for task 3

```
Task 3. Hold out 2017-2018. Compare
seasonal naive, Holt-Winters, SARIMA and
two regressions by MAE, RMSE and MAPE
in a great_tables table.
```

Prompt for task 4

```
Task 4. Forecast 2019 with the best
method and with SARIMA, 95 % bands.
Which band would you trust for stock
planning, and why?
```

Check every number against your own code: you own the result.

Open data for forecasting

ABS Retail Trade

Monthly turnover in Australia since 1982, by state and industry

Licence: CC BY 4.0

Eurostat

Monthly retail trade and tourism nights by EU country

Licence: free reuse, cite the source

M5 (Walmart)

Daily unit sales of 3,049 products in 10 US stores, 2011–2016

Licence: competition data: check the terms

The licence decides: commit only data you may share; link the rest.

Links and loading code: `forecasting.qmd` (appendix)

Sources: ABS, Eurostat and M5 dataset pages (links in forecasting.qmd)

Key takeaways

1. **Plot and smooth first:** trend, season, cycle; additive or multiplicative; stationary or not?
2. **Start simple:** naive, drift, averages, smoothing and a trend line: all on a calculator
3. **Test on the future:** hold out the last periods; compare MAE, RMSE and MAPE
4. **Beat the simplest benchmark:** on seasonal data, the seasonal naive
5. **Report the band, check the errors:** autocorrelated errors make a band too narrow

Next: run sessions/01-foundations/forecasting.qmd; it reproduces the results on the slides.

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Australian Bureau of Statistics (2019). Retail Trade, Australia, April 2019, cat. no. 8501.0, Table 11. CC BY 4.0.

Seabold, S. & Perktold, J. (2010). statsmodels: Econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference.

Free textbook: `otexts.com/fpp3` · Python edition: `otexts.com/fpppy`

Thank you. Questions and discussion

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